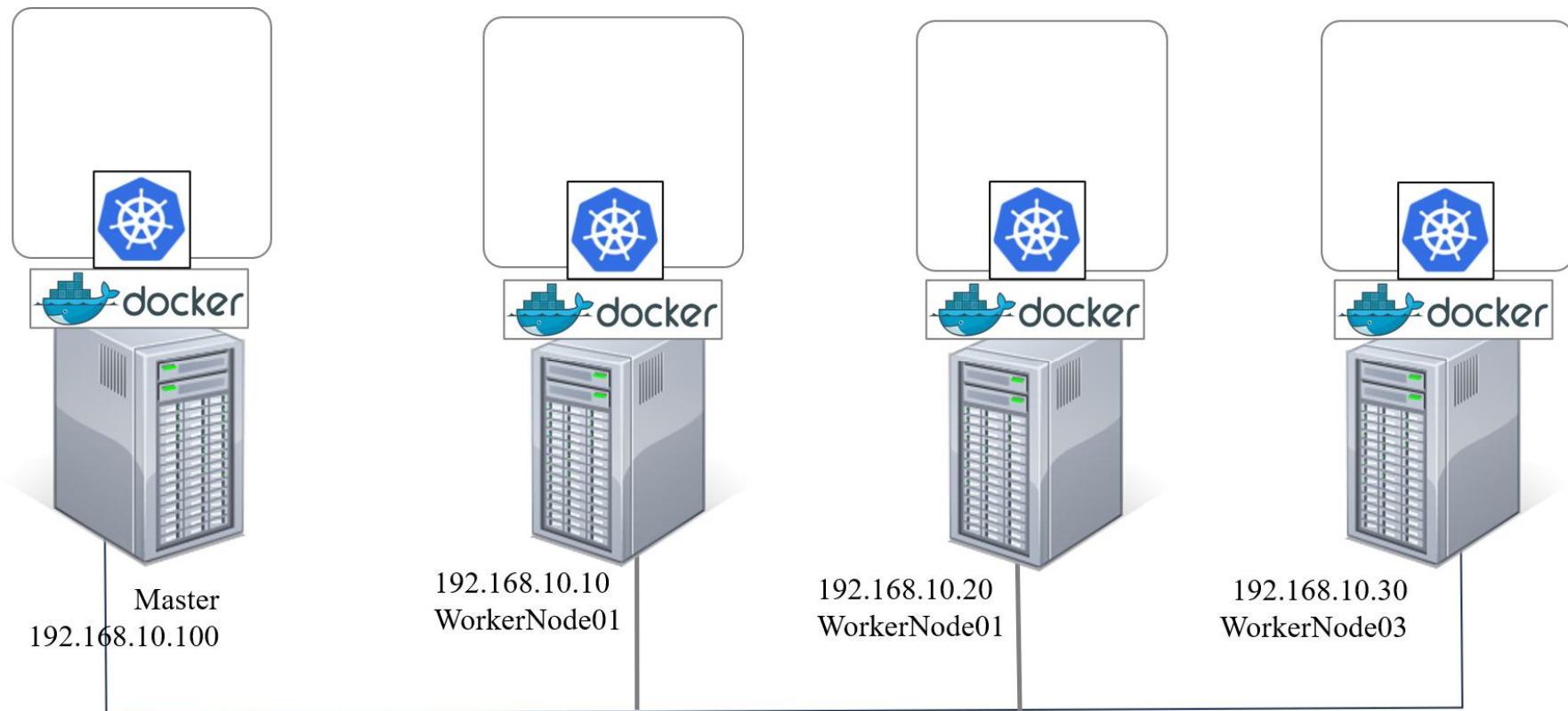


K8S Cluster Setup



1. [All nodes] 설치 전 환경 설정

1) 타임존 설정

```
timedatectl set-timezone Asia/Seoul
```

2) 방화벽 해제

```
systemctl stop ufw
```

```
systemctl disable ufw
```

```
ufw disable
```

2) Swap 비활성화

```
swapoff -a
```

```
nano /etc/fstab ==> SWAP 관련 설정 모두 비활성화
```

3) 필수 패키지 설치

```
apt-get update
```

```
apt-get install -y apt-transport-https ca-certificates curl gpg iproute2
```

2. [All nodes] 컨테이너 런타임 설치 전 사전작업

1) 필요한 모듈 로드

```
cat <<EOF | tee /etc/modules-load.d/containerd.conf
```

```
overlay
```

```
br_netfilter
```

```
EOF
```

```
modprobe overlay
```

```
modprobe br_netfilter
```

2) iptables 설정

```
tee /etc/sysctl.d/kubernetes.conf <<EOF
```

```
net.bridge.bridge-nf-call-ip6tables = 1
```

```
net.bridge.bridge-nf-call-iptables = 1
```

```
net.ipv4.ip_forward = 1
```

```
EOF
```

3) 설정값 적용

```
sysctl --system
```

3. [All nodes] 컨테이너 런타임 설치

1) 도커 공식 GPG 키 추가

```
apt-get update
```

```
install -m 0755 -d /etc/apt/keyrings
```

```
curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
```

```
chmod a+r /etc/apt/keyrings/docker.asc
```

2) 도커 Apt 저장소 추가

```
echo ₩
```

```
"deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc] https://download.docker.com/linux/ubuntu ₩  
$(. /etc/os-release && echo "${UBUNTU_CODENAME:-$VERSION_CODENAME}") stable" | ₩  
tee /etc/apt/sources.list.d/docker.list > /dev/null
```

3) containerd 설치

```
apt-get update
```

```
apt-get install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
```

4) CRI 활성화

```
mkdir -p /etc/containerd
```

```
containerd config default | tee /etc/containerd/config.toml
```

```
sed -i 's/SystemdCgroup = false/SystemdCgroup = true/g' /etc/containerd/config.toml
```

```
systemctl restart containerd
```

4. [All nodes] K8S 설치

1) K8S GPG key와 저장소 지정

```
apt-get install curl ca-certificates apt-transport-https -y
curl -fsSL https://pkgs.k8s.io/core:/stable:/v1.31/deb/Release.key | gpg --dearmor -o /etc/apt/keyrings/kubernetes-apt-keyring.gpg
echo "deb [signed-by=/etc/apt/keyrings/kubernetes-apt-keyring.gpg] https://pkgs.k8s.io/core:/stable:/v1.31/deb/ /" | tee
/etc/apt/sources.list.d/kubernetes.list
chmod a+r /etc/apt/keyrings/kubernetes-apt-keyring.gpg
```

*Docker 공식 공개키 등록하는 과정 : 공개키(GPG key)를 가져와 시스템에 등록 후 Docker의 GPG 공개키를 읽을 수 있도록 권한 변경

2) Kubeadm, kubectl, kubelet 설치

```
apt update
apt install kubelet kubeadm kubectl -y
apt-mark hold kubelet kubeadm kubectl
```

*kubeadm	Kubernetes 클러스터 설치 및 초기 구성 도구
*kubelet	각 노드에서 Pod를 실제로 실행·관리하는 에이전트
*kubectl	클러스터를 제어하는 사용자 명령 인터페이스(CLI)

[오류메세지] E: Could not get lock /var/lib/dpkg/lock-frontent. It is held by process ~~

kill -9 PID

dpkg --configure -a

3) K8S 서비스 활성화

`systemctl daemon-reload`

`systemctl restart kubelet`

`systemctl enable kubelet`

5. [Master] Kubeadm을 이용한 cluster 구성

1) 마스터 노드 초기화

```
kubeadm init --pod-network-cidr=192.168.0.0/16
```

<<이전 설정값으로 초기화 시 >>

```
kubeadm reset -f
```

```
rm -rf /etc/Kubernetes
```

```
rm -rf /var/lib/etcd
```

```
rm -rf $HOME/.kube
```

```
rm -f /etc/kubernetes/pki/ca.crt
```

2) Kubernetes 관리자 계정(kubectl)이 클러스터에 접근할 수 있도록 인증 설정을 연결하는 과정

```
mkdir -p $HOME/.kube
```

```
cp -i /etc/kubernetes/admin.conf $HOME/.kube/config
```

```
chown $(id -u):$(id -g) $HOME/.kube/config
```

<< 생성된 토큰 값 확인 >>

```
kubeadm join 192.168.10.10:6443 --token mxh567.vq5b67pvo30kh1sc ₩
```

```
    --discovery-token-ca-cert-hash sha256:713abf30b0502dd416ee44cc8e8a674d8e37b2b4114c515f392595d1cf6de58d
```

<토큰 재발급> kubeadm token list master ==> kubeadm token create --print-join-command

3) CNI "Calico(캘리코)" 설치 ** CNI(Container Network Interface) : Pod 간 통신이 가능하게 하는 인터페이스(표준 사양)

```
kubectl create -f https://raw.githubusercontent.com/projectcalico/calico/v3.28.0/manifests/tigera-operator.yaml
curl https://raw.githubusercontent.com/projectcalico/calico/v3.28.0/manifests/custom-resources.yaml -O
kubectl create -f custom-resources.yaml
```

<<확인명령어 1>>

```
root@master:~# kubectl get pods -n kube-system
```

NAME	READY	STATUS	RESTARTS	AGE
calico-node-dbjbw	0/1	Running	0	40s

<<확인명령어 2>>

```
root@master:~# kubectl get ns
```

NAME	STATUS	AGE
calico-apiserver	Active	4d7h
calico-system	Active	4d7h

<<확인명령어 3>>

```
root@master:~# kubectl get nodes
```

NAME	STATUS	ROLES	AGE	VERSION
master	Ready	control-plane	27m	v1.31.13

6. [Worker Node] Master node와 Worker node Join (Cluster 구성)

```
apt-get update && apt-get install -y conntrack
```

```
kubeadm join 192.168.10.10:6443 --token mxh567.vq5b67pvo30kh1sc ₩
```

```
--discovery-token-ca-cert-hash sha256:713abf30b0502dd416ee44cc8e8a674d8e37b2b4114c515f392595d1cf6de58d
```

<<설정 초기화 시 >>

```
kubeadm reset -f
```

```
rm -rf /etc/Kubernetes
```

```
rm -rf /var/lib/etcd
```

```
rm -rf $HOME/.kube
```

```
rm -f /etc/kubernetes/pki/ca.crt
```

7. [Master] K8S Cluster 확인

```
kubectl get nodes
```